

PROPORTIONS OF THE NON-TRIVIAL ZEROS OF THE RIEMANN ZETA FUNCTION

BIAO WANG

ABSTRACT. Let $C_0 = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot\left(\frac{1}{\sqrt{2}}\right) = 0.67250\dots$ and $C_1 = \frac{C_0+1}{2} = 0.83625\dots$. Recently, it is obtained by Alpöge and Furman that more than 67.25% of the non-trivial zeros of the Riemann zeta function are simple and on the critical line, and more than 83.62% are distinct. Later, Lamzouri gave a different and more direct proof. In this article, by refining the method of Lamzouri, we slightly improve the bounds C_0 and C_1 in these two results to $C_0 + \delta_0$ and $C_1 + \frac{\delta_0}{2}$ with $\delta_0 = 6.66624\dots \times 10^{-8}$, respectively.

1. INTRODUCTION

Let $\zeta(s)$ be the Riemann zeta function. In 1859, Riemann asserted that all the non-trivial zeros of $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. In 1914, Hardy [4] proved that there are infinitely many, and in 1942 Selberg [12] established a positive proportion. In 1956, Min [8] gave an explicit value. In 1974, Levinson [7] introduced a new mollifier method and proved that more than one third lie on the line. In 1989, Conrey [3] raised this proportion to more than two fifths. Subsequent developments of the mollifier method include the results of Feng [5] and Pratt, Robles, Zaharescu and Zeindler [11].

Besides the mollifier method, a complementary approach uses pair correlation and inequalities for finite collections of zeros. In 1973, Montgomery [9] initiated this approach and proved, under the Riemann hypothesis, that at least two thirds of the zeros are simple. Montgomery and Taylor [10] subsequently obtained a larger proportion C_0 defined by (1.1) below. Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [2] established an unconditional pair-correlation formula. Recently, this formula was used to obtain an unconditional proof of Montgomery's result by Alpöge and Furman [1]. Later, Lamzouri [6] gave a different and more direct proof. In the proof, he established an inequality on the number of real simple elements in any finite multiset invariant under complex conjugation. In this article, we will refine Lamzouri's inequality and slightly improve the Montgomery-Taylor's bound C_0 on the simple critical zeros of the Riemann zeta function. We state our main result as follows. For the reader's convenience, we use Lamzouri's notation throughout.

A nontrivial zero of $\zeta(s)$ is written $\rho = \beta + i\gamma$, $0 \leq \beta \leq 1$, and its multiplicity is m_ρ . Put

$$N(T) := \sum_{0 < \gamma \leq T}^{\rho} 1, \quad N_0^s(T) := |\{\rho : 0 < \gamma \leq T, \beta = \tfrac{1}{2}, m_\rho = 1\}|,$$

and $N_d(T) := |\{\rho : 0 < \gamma \leq T\}|$, which counts distinct zeros. Unless otherwise specified, sums over zeros or multisets count multiplicity. Let

$$(1.1) \quad C_0 := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot\left(\frac{1}{\sqrt{2}}\right) = 0.67250\dots,$$

$$(1.2) \quad C_1 := \frac{C_0 + 1}{2} = 0.83625\dots$$

Date: September 22, 2026.

2020 Mathematics Subject Classification. Primary 11M06, 11M26.

Key words and phrases. Riemann zeta-function, non-trivial zeros, multisets.

For the explicit constants in our result, let

$$(1.3) \quad H_0 := 372018941724 \times 10^{-11}, \quad a_0 := \frac{2(\sqrt{5} - 2)^2}{3(1 + 2\pi^2 H_0^2)^2}, \quad \delta_0 := \frac{a_0(C_0 - 2/H_0)}{1 - a_0}.$$

In particular,

$$\delta_0 = 6.66624 \dots \times 10^{-8} > 0.$$

Theorem 1.1. *We have, unconditionally,*

$$(1.4) \quad \liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq C_0 + \delta_0,$$

and

$$(1.5) \quad \liminf_{T \rightarrow \infty} \frac{N_d(T)}{N(T)} \geq C_1 + \frac{\delta_0}{2}.$$

Both in Theorem A of Alpöge and Furman [1] and Theorem 1.1 of Lamzouri [6], the lower bounds in (1.4) and (1.5) are C_0 and C_1 respectively. Hence Theorem 1.1 gives slightly larger lower bounds. The improvement rests on two ingredients. First, in Section 2 we will refine Lamzouri's inequality on finite-multisets of conjugation invariance by retaining an extra spectral term $\Delta_K(\mathcal{Z})$. This term yields the gains δ_0 and $\delta_0/2$ in (1.4) and (1.5), respectively. Second, to estimate the spectral term, in Section 3 we will divide the simple real elements in a multiset into triples and give a quantitative three-point kernel bound. Finally, in Section 4, we will use the refined Lamzouri's inequality and the estimate of $\Delta_K(\mathcal{Z})$ to complete the proof of Theorem 1.1.

2. A REFINEMENT OF THE FINITE-MULTISET INEQUALITY

For compactly supported $f \in L^1(\mathbb{R})$, we define the Fourier transform of f by

$$\widehat{f}(\xi) := \int_{\mathbb{R}} f(u) e^{-2\pi i \xi u} du, \quad \xi \in \mathbb{C}.$$

Let $\mathcal{Z}$ be a finite multiset. For $z \in \mathcal{Z}$, let m_z denote its multiplicity. If $m_z = 1$, then z is called a *simple point*. We say that $\mathcal{Z}$ is *invariant under complex conjugation* or is of *conjugation invariance* if $\bar{z} \in \mathcal{Z}$ and $m_{\bar{z}} = m_z$ whenever $z \in \mathcal{Z}$. In [6, Proposition 2.1], Lamzouri established an inequality on the number of real simple elements in such multisets. In the following, we give a refinement of Lamzouri's inequality by introducing an extra summand $\Delta_K(\mathcal{Z})$. Omitting $\Delta_K(\mathcal{Z})$ from (2.1) and (2.2) recovers inequalities (2.4) and (2.5) of [6, Proposition 2.1], respectively.

Let A be an $n \times n$ Hermitian and positive semidefinite complex matrix. Then it admits a spectral decomposition

$$A = U \operatorname{diag}(\lambda_1, \dots, \lambda_n) U^*, \quad \lambda_j \geq 0,$$

where U is unitary. Define

$$\Psi(t) := \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2, \end{cases}$$

and

$$\Psi(A) := U \operatorname{diag}(\Psi(\lambda_1), \dots, \Psi(\lambda_n)) U^*.$$

This definition is independent of the choice of orthonormal eigenbasis. In particular,

$$\operatorname{tr} \Psi(A) = \sum_{j=1}^n \Psi(\lambda_j) \geq 0.$$

Proposition 2.1. *Let $\lambda > 0$ be a real number and let $\eta \in L^2(\mathbb{R})$ be a real valued even function with $\operatorname{supp}(\eta) \subset (-\lambda, \lambda)$, such that $\widehat{\eta^2}(0) = 1$. Let $\mathcal{Z}$ be a non-empty finite multiset of complex numbers which is invariant under complex conjugation. Let $\mathcal{R}_1 = \{x_1, \dots, x_n\}$ be the set of simple real elements of $\mathcal{Z}$. Let $K(\xi) = \widehat{\eta^2}(\xi)$. Define*

$$G_K := (K(x_j - x_\ell))_{1 \leq j, \ell \leq n}, \quad \Delta_K(\mathcal{Z}) := \operatorname{tr} \Psi(G_K).$$

If $n = 0$, set $\Delta_K(\mathcal{Z}) = 0$. Then we have

$$(2.1) \quad \sum_{\substack{z \in \mathcal{Z} \cap \mathbb{R} \\ m_z=1}} 1 \geq 2 \sum_{z \in \mathcal{Z}} 1 - \sum_{z, s \in \mathcal{Z}} K(z-s)^2 + \Delta_K(\mathcal{Z}).$$

Moreover, the number of distinct elements of $\mathcal{Z}$ is at least

$$(2.2) \quad \frac{3}{2} \sum_{z \in \mathcal{Z}} 1 - \frac{1}{2} \sum_{z, s \in \mathcal{Z}} K(z-s)^2 + \frac{1}{2} \Delta_K(\mathcal{Z}).$$

Here $\Delta_K(\mathcal{Z}) \geq 0$, and every sum over $\mathcal{Z}$ counts multiplicity.

Proof. For $z \in \mathbb{C}$ define

$$f_z(u) := \eta(u)e^{-2\pi iuz}, \quad g_z(u) := \frac{f_z(u) + f_{\bar{z}}(u)}{2}, \quad h_z(u) := \frac{f_z(u) - f_{\bar{z}}(u)}{2i}.$$

Define the inner product on $L^2((-\lambda, \lambda), \mathbb{C})$, the space of complex-valued square-integrable functions on the interval $(-\lambda, \lambda)$, by $\langle f, g \rangle = \int_{-\lambda}^{\lambda} f(u)\overline{g(u)} du$. Then by [6, (2.9)-(2.11)], we have

$$(2.3) \quad \langle f_z, f_s \rangle = K(z - \bar{s}), \quad \|f_x\|^2 = 1 \ (x \in \mathbb{R}), \quad \|g_z\|^2 - \|h_z\|^2 = 1.$$

Let $\mathcal{R}$ be the set of distinct real elements of $\mathcal{Z}$ and $\mathcal{S}$ the set of its distinct non-real elements. Split $\mathcal{R}$ into its simple and multiple elements, denoted by $\mathcal{R}_1$ and $\mathcal{R}_2$:

$$\mathcal{R}_1 = \{x_1, \dots, x_n\}, \quad \mathcal{R}_2 = \{x_{n+1}, \dots, x_{n+r}\}, \quad \mathcal{S} = \{z_1, \bar{z}_1, \dots, z_k, \bar{z}_k\}.$$

Thus the number of distinct elements is $n + r + 2k$. For brevity within this proof only, write

$$\mathcal{N} := \sum_{z \in \mathcal{Z}} 1, \quad \mathcal{E}_K := \sum_{z, s \in \mathcal{Z}} K(z-s)^2.$$

On the finite-dimensional space

$$W = \text{Span}_{\mathbb{C}}(f_{x_1}, \dots, f_{x_{n+r}}, g_{z_1}, \dots, g_{z_k}, h_{z_1}, \dots, h_{z_k}),$$

we introduce the self-adjoint operator

$$A = \sum_{\ell=1}^{n+r} m_{x_\ell} f_{x_\ell} \otimes f_{x_\ell} + 2 \sum_{\ell=1}^k m_{z_\ell} (g_{z_\ell} \otimes g_{z_\ell} - h_{z_\ell} \otimes h_{z_\ell}).$$

Here $v \otimes w$ maps ξ to $\langle \xi, w \rangle v$. Conjugation invariance of $\mathcal{Z}$ then gives

$$A = \sum_{z \in \mathcal{R} \cup \mathcal{S}} m_z f_z \otimes f_{\bar{z}}.$$

Notice that for rank-one operators $v \otimes w$, we have

$$\text{tr}(v \otimes w) = \langle v, w \rangle, \quad \text{tr}((v \otimes w)(v' \otimes w')) = \langle v', w \rangle \langle v, w' \rangle.$$

Consequently, by (2.3) each term of $\text{tr} A$ equals $m_z K(0) = m_z$, and the (z, s) term of $\text{tr}(A^2)$ equals $m_z m_s K(s - z) K(z - s) = m_z m_s K(z - s)^2$. Here evenness of η^2 implies evenness of K . Thus, we obtain

$$(2.4) \quad \text{tr} A = \mathcal{N}, \quad \text{tr}(A^2) = \mathcal{E}_K.$$

In particular $\mathcal{E}_K$ is real and nonnegative, since A is self-adjoint.

Put $P = \sum_{\ell=1}^n f_{x_\ell} \otimes f_{x_\ell}$ and $Q = A - P$. Since $m_{x_\ell} = 1$ for $1 \leq \ell \leq n$, we can write $Q = B - C$, where

$$B = \sum_{\ell=n+1}^{n+r} m_{x_\ell} f_{x_\ell} \otimes f_{x_\ell} + 2 \sum_{\ell=1}^k m_{z_\ell} g_{z_\ell} \otimes g_{z_\ell}, \quad C = 2 \sum_{\ell=1}^k m_{z_\ell} h_{z_\ell} \otimes h_{z_\ell}.$$

Both operators B and C are positive semidefinite and $\text{rank } B \leq r + k$. If the positive spectral eigenspace of Q had dimension larger than $r + k$, it would contain a nonzero vector ξ orthogonal to the range of B . But then $\langle Q\xi, \xi \rangle = -\langle C\xi, \xi \rangle \leq 0$, contradicting positivity on that eigenspace. Thus, in the spectral decomposition $Q = Q_+ - Q_-$, where Q_+ and Q_- denote respectively

the positive and negative parts of Q , we have $\text{rank } Q_+ \leq r + k$. Moreover, $P, Q_\pm$ are positive semidefinite and $Q_+Q_- = Q_-Q_+ = 0$.

Since $A = P + Q = P + Q_+ - Q_-$, expanding the square of A gives

$$A^2 = P^2 + Q_+^2 + Q_-^2 + PQ_+ + Q_+P - PQ_- - Q_-P - Q_+Q_- - Q_-Q_+.$$

By $Q_+Q_- = Q_-Q_+ = 0$, the commutativity of the trace yields

$$\mathcal{E}_K = \text{tr}(P^2) + \text{tr}(Q_+^2) + \text{tr}(Q_-^2) + 2\text{tr}(PQ_+) - 2\text{tr}(PQ_-).$$

Since P is positive semidefinite, we have $\text{tr}(PQ_+) \geq 0$. Discarding this nonnegative term therefore gives

$$(2.5) \quad \mathcal{E}_K \geq \text{tr}(P^2) + \text{tr}(Q_+^2) + \text{tr}(Q_-^2) - 2\text{tr}(PQ_-).$$

Let $d_W = \dim W$, and let

$$p_1 \geq \cdots \geq p_{d_W} \geq 0, \quad \nu_1 \geq \cdots \geq \nu_{d_W} \geq 0$$

be the eigenvalues of P and Q_- , respectively, counted with multiplicity. Then

$$\text{tr}(P^2) = \sum_{j=1}^{d_W} p_j^2, \quad \text{tr}(Q_-^2) = \sum_{j=1}^{d_W} \nu_j^2.$$

Let $\{e_j\}_{j=1}^{d_W}$ be an orthonormal eigenbasis of Q_- , and let $\{w_i\}_{i=1}^{d_W}$ be an orthonormal eigenbasis of P , with

$$Q_-e_j = \nu_j e_j, \quad Pw_i = p_i w_i, \quad d_j := \langle Pe_j, e_j \rangle.$$

Then

$$e_j = \sum_{i=1}^{d_W} \langle e_j, w_i \rangle w_i, \quad Pe_j = \sum_{i=1}^{d_W} p_i \langle e_j, w_i \rangle w_i,$$

and

$$d_j = \langle Pe_j, e_j \rangle = \sum_{i=1}^{d_W} p_i \langle e_j, w_i \rangle \langle w_i, e_j \rangle = \sum_{i=1}^{d_W} p_i |\langle e_j, w_i \rangle|^2.$$

For $1 \leq \ell \leq d_W$, define

$$\theta_i := \sum_{j=1}^{\ell} |\langle e_j, w_i \rangle|^2.$$

Geometrically, θ_i is the squared norm of the orthogonal projection of w_i onto $\text{span}\{e_1, \dots, e_\ell\}$. Hence, by Parseval's identity we have $0 \leq \theta_i \leq 1$. Summing the preceding identity over $j = 1, \dots, \ell$ yields

$$\sum_{j=1}^{\ell} d_j = \sum_{i=1}^{d_W} p_i \theta_i.$$

Applying Parseval's identity to the orthonormal basis $\{w_i\}$ also gives

$$\sum_{i=1}^{d_W} \theta_i = \sum_{j=1}^{\ell} \sum_{i=1}^{d_W} |\langle e_j, w_i \rangle|^2 = \sum_{j=1}^{\ell} \|e_j\|^2 = \ell.$$

It follows that

$$\sum_{i=1}^{\ell} (1 - \theta_i) = \sum_{i=\ell+1}^{d_W} \theta_i,$$

and since the p_i decrease, we get that

$$\sum_{i=1}^{\ell} p_i - \sum_{i=1}^{d_W} p_i \theta_i = \sum_{i=1}^{\ell} p_i (1 - \theta_i) - \sum_{i=\ell+1}^{d_W} p_i \theta_i \geq p_\ell \sum_{i=1}^{\ell} (1 - \theta_i) - p_\ell \sum_{i=\ell+1}^{d_W} \theta_i = 0.$$

This implies that

$$\sum_{j=1}^{\ell} d_j \leq \sum_{i=1}^{\ell} p_i.$$

Now, by $Q_- e_j = \nu_j e_j$ and $d_j := \langle P e_j, e_j \rangle$, we have

$$\text{tr}(PQ_-) = \sum_{j=1}^{d_W} \langle PQ_- e_j, e_j \rangle = \sum_{j=1}^{d_W} \nu_j \langle P e_j, e_j \rangle = \sum_{j=1}^{d_W} \nu_j d_j.$$

Set $\nu_{d_W+1} = 0$. Summation by parts now gives

$$\text{tr}(PQ_-) = \sum_{\ell=1}^{d_W} (\nu_{\ell} - \nu_{\ell+1}) \sum_{j=1}^{\ell} d_j \leq \sum_{\ell=1}^{d_W} (\nu_{\ell} - \nu_{\ell+1}) \sum_{j=1}^{\ell} p_j = \sum_{j=1}^{d_W} \nu_j p_j.$$

Therefore by (2.5), we obtain that

$$(2.6) \quad \mathcal{E}_K \geq \text{tr}(Q_+^2) + \sum_{j=1}^{d_W} (p_j - \nu_j)^2.$$

Using $a^2 \geq 4a - 4$ on the at most $r + k$ positive eigenvalues,

$$\text{tr}(Q_+^2) \geq 4 \text{tr} Q_+ - 4(r + k) = 4 \text{tr} Q + 4 \sum_{j=1}^{d_W} \nu_j - 4(r + k).$$

For $p \geq 0$, we have

$$\Phi(p) := \min_{\nu \geq 0} \{(p - \nu)^2 + 4\nu\} = \begin{cases} p^2, & 0 \leq p \leq 2, \\ 4p - 4, & p \geq 2, \end{cases} = 2p - 1 + \Psi(p).$$

In particular $\Phi(0) = 0$. By (2.6), we get that

$$(2.7) \quad \mathcal{E}_K \geq 4 \text{tr} Q + \sum_{j=1}^{d_W} \Phi(p_j) - 4(r + k).$$

Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be the standard coordinate basis of $\mathbb{C}^n$, and define a linear operator $V : \mathbb{C}^n \rightarrow W$ by $V \mathbf{e}_j = f_{x_j}$. Then for $c = (c_1, \dots, c_n) \in \mathbb{C}^n$,

$$Vc = \sum_{j=1}^n c_j f_{x_j}.$$

For any $\xi \in W$, we have

$$\langle Vc, \xi \rangle_W = \sum_{j=1}^n c_j \langle f_{x_j}, \xi \rangle_W = \left\langle c, (\langle \xi, f_{x_j} \rangle_W)_{j=1}^n \right\rangle_{\mathbb{C}^n}.$$

Consequently, the adjoint of V is given by

$$V^* \xi = (\langle \xi, f_{x_j} \rangle_W)_{j=1}^n.$$

It follows that

$$VV^* \xi = \sum_{j=1}^n \langle \xi, f_{x_j} \rangle f_{x_j} = \left(\sum_{j=1}^n f_{x_j} \otimes f_{x_j} \right) \xi = P \xi.$$

We obtain $VV^* = P$.

On the other hand, for $c \in \mathbb{C}^n$,

$$(V^* Vc)_j = \langle Vc, f_{x_j} \rangle = \sum_{\ell=1}^n c_{\ell} \langle f_{x_{\ell}}, f_{x_j} \rangle.$$

Hence the (j, ℓ) entry of the matrix of V^*V in the standard coordinate basis is

$$(V^*V)_{j\ell} = \langle f_{x_\ell}, f_{x_j} \rangle = K(x_\ell - x_j) = K(x_j - x_\ell).$$

Thus, $V^*V = G_K$.

Notice that nonzero eigenvalues of V^*V and VV^* coincide, including multiplicities. Since $\Phi(0) = 0$ and

$$\text{tr } G_K = \sum_{j=1}^n (G_K)_{jj} = \sum_{j=1}^n K(x_j - x_j) = \sum_{j=1}^n K(0) = \sum_{j=1}^n 1 = n,$$

this proves

$$\sum_{j=1}^{d_W} \Phi(p_j) = \text{tr } \Phi(G_K) = 2 \text{tr } G_K - n + \text{tr } \Psi(G_K) = n + \Delta_K(\mathcal{Z}).$$

When $n = 0$, both sides are zero by convention. Combining (2.7) with $\text{tr } Q = \mathcal{N} - n$ gives

$$(2.8) \quad \mathcal{E}_K \geq 4\mathcal{N} - 3n - 4(r + k) + \Delta_K(\mathcal{Z}).$$

Finally, using the trivial estimate $\mathcal{N} - n \geq 2r + 2k$ in (2.8) implies (2.1). Let $\mathcal{D}$ denote the number of distinct elements of $\mathcal{Z}$, then we have $\mathcal{D} = n + r + 2k$. Using (2.8) again, by

$$(4\mathcal{N} - 3n - 4r - 4k) - (3\mathcal{N} - 2(n + r + 2k)) = \mathcal{N} - n - 2r \geq 2k \geq 0.$$

we have

$$\mathcal{E}_K \geq 3\mathcal{N} - 2(n + r + 2k) + \Delta_K(\mathcal{Z}) = 3\mathcal{N} - 2\mathcal{D} + \Delta_K(\mathcal{Z}).$$

Rearranging gives

$$\mathcal{D} \geq \frac{3}{2}\mathcal{N} - \frac{1}{2}\mathcal{E}_K + \frac{1}{2}\Delta_K(\mathcal{Z}),$$

which proves (2.2).

The matrix G_K is a Hermitian and positive semidefinite Gram matrix, so $\Delta_K(\mathcal{Z}) \geq 0$. $\square$

3. A QUANTITATIVE THREE-POINT KERNEL BOUND

Lemma 3.1. *Under the hypotheses of Proposition 2.1, suppose $\mathcal{T}_1, \dots, \mathcal{T}_J$ are disjoint triples in $\mathcal{R}_1$ such that*

$$\sum_{\substack{x, y \in \mathcal{T}_j \\ x < y}} K(x - y)^2 \geq e_*, \quad 1 \leq j \leq J, \quad 0 \leq e_* \leq \frac{1}{2}.$$

*Then $\Delta_K(\mathcal{Z}) \geq 2e_*J$.*

Proof. For one 3×3 principal Gram block B , it is a positive semidefinite Gram matrix. Let $\lambda_1, \lambda_2, \lambda_3$ be the three eigenvalues of B . Then $\lambda_1, \lambda_2, \lambda_3 \geq 0$ and $\text{tr } B = K(0) + K(0) + K(0) = 3$.

If $0 \leq \lambda_1, \lambda_2, \lambda_3 \leq 2$, then on this interval, $\Psi(t) = (t - 1)^2$. Writing

$$B = U \text{diag}(\lambda_1, \lambda_2, \lambda_3) U^*,$$

where U is unitary, then

$$\Psi(B) = U \text{diag}((\lambda_1 - 1)^2, (\lambda_2 - 1)^2, (\lambda_3 - 1)^2) U^* = (B - I_3)^2.$$

Consequently, $\text{tr } \Psi(B) = \text{tr}((B - I_3)^2)$.

Set $C = B - I_3$. Since every diagonal entry of B equals one, we have $C_{ii} = 0$ and $C_{ij} = B_{ij}$ for $i \neq j$. Moreover, C is Hermitian, so $C_{ji} = \overline{C_{ij}}$. Therefore,

$$\text{tr}(C^2) = \sum_{i=1}^3 (C^2)_{ii} = \sum_{i=1}^3 \sum_{j=1}^3 C_{ij} C_{ji} = \sum_{i,j=1}^3 |C_{ij}|^2 = \sum_{i \neq j} |B_{ij}|^2 = 2 \sum_{i < j} |B_{ij}|^2.$$

If B corresponds to the triple $\mathcal{T} = \{x_a, x_b, x_c\}$, then, since K is real-valued on the real axis,

$$\sum_{i < j} |B_{ij}|^2 = K(x_a - x_b)^2 + K(x_a - x_c)^2 + K(x_b - x_c)^2 \geq e_*,$$

by the hypothesis on $\mathcal{T}$. Combining these identities yields

$$\operatorname{tr} \Psi(B) = \operatorname{tr}((B - I_3)^2) = 2 \sum_{i < j} |B_{ij}|^2 \geq 2e_*.$$

If $\lambda_i > 2$ for some $1 \leq i \leq 3$, then $\Psi(\lambda_i) = 2\lambda_i - 3 > 1$. Since $\Psi(\lambda_j) \geq 0$ for all $1 \leq j \leq 3, j \neq i$, and $0 \leq e_* \leq 1/2$, it follows that

$$\operatorname{tr} \Psi(B) \geq \Psi(\lambda_i) > 1 \geq 2e_*.$$

Thus, we always have

$$(3.1) \quad \operatorname{tr} \Psi(B) \geq 2e_*$$

for each principal block B corresponding to one of the triples $\mathcal{T}_1, \dots, \mathcal{T}_J$.

Now, let $I_1, \dots, I_J$ be the mutually disjoint index sets corresponding to the triples $\mathcal{T}_1, \dots, \mathcal{T}_J$, and let I_0 consist of the remaining indices. Define the coordinate subspaces

$$E_a := \operatorname{span}\{\mathbf{e}_i : i \in I_a\}, \quad 0 \leq a \leq J,$$

where $\mathbf{e}_1, \dots, \mathbf{e}_n$ are the standard coordinate vectors of $\mathbb{C}^n$. These subspaces give the orthogonal decomposition

$$\mathbb{C}^n = E_0 \oplus E_1 \oplus \dots \oplus E_J.$$

If I_0 is empty, the corresponding subspace and block are omitted.

Let $0 \leq a \leq J$. Let Π_a denote the orthogonal projection onto E_a . The principal block indexed by I_a may be written as

$$B_a = \Pi_a G_K \Pi_a \Big|_{E_a}.$$

Choose an orthonormal eigenbasis for each B_a . Viewed as vectors in $\mathbb{C}^n$ by extending their coordinates by zero outside I_a , these bases together form an orthonormal basis $u_1, \dots, u_n$ of $\mathbb{C}^n$.

Suppose that $u_j \in E_a$ and that $B_a u_j = \mu_j u_j$. Since $\Pi_a u_j = u_j$, $\Pi_a^* = \Pi_a$, and $\|u_j\| = 1$, we have

$$\langle G_K u_j, u_j \rangle = \langle G_K u_j, \Pi_a u_j \rangle = \langle \Pi_a G_K u_j, u_j \rangle = \langle B_a u_j, u_j \rangle = \mu_j.$$

Thus μ_j is the j th diagonal entry of G_K in the basis $\{u_j\}$, although u_j need not be an eigenvector of G_K . Let $w_1, \dots, w_n$ be an orthonormal eigenbasis of G_K , with $G_K w_i = \lambda_i w_i$. Expanding u_j in this basis, we obtain

$$u_j = \sum_{i=1}^n \langle u_j, w_i \rangle w_i,$$

and hence

$$\mu_j = \langle G_K u_j, u_j \rangle = \sum_{i=1}^n \lambda_i |\langle u_j, w_i \rangle|^2.$$

By using Parseval's identity for the orthonormal basis $\{w_i\}$, the coefficients satisfy

$$|\langle u_j, w_i \rangle|^2 \geq 0, \quad \sum_{i=1}^n |\langle u_j, w_i \rangle|^2 = \|u_j\|^2 = 1.$$

Therefore, each μ_j is a convex combination of the eigenvalues of G_K .

By the convexity of Ψ , Jensen's inequality gives

$$\Psi(\mu_j) \leq \sum_{i=1}^n |\langle u_j, w_i \rangle|^2 \Psi(\lambda_i).$$

Summing over j and using Parseval's identity for the orthonormal basis $\{u_j\}$ gives

$$\sum_{a=0}^J \operatorname{tr} \Psi(B_a) = \sum_{j=1}^n \Psi(\mu_j) \leq \sum_{i=1}^n \Psi(\lambda_i) \sum_{j=1}^n |\langle u_j, w_i \rangle|^2 = \sum_{i=1}^n \Psi(\lambda_i) = \operatorname{tr} \Psi(G_K).$$

If $I_0 = \emptyset$, we interpret the contribution of $\operatorname{tr} \Psi(B_0)$ as zero. By (3.1) and $\operatorname{tr} \Psi(B_0) \geq 0$, we conclude that

$$\Delta_K(\mathcal{Z}) = \operatorname{tr} \Psi(G_K) \geq 2e_* J,$$

as desired. $\square$

Let

$$f_0(u) = \frac{\cos(\sqrt{2}u)}{\sqrt{2}\sin(1/\sqrt{2})} \mathbf{1}_{[-1/2, 1/2]}(u), \quad K_0 = \widehat{f_0},$$

which is used by Lamzouri in the proof of [6, Lemma 3.2]. Then f_0 is nonnegative, even, and has integral one. In the following, we give a quantitative three-point kernel bound.

Lemma 3.2. *For every $H > 0$ and $u, v \geq 0$ with $u + v \leq H$, we have*

$$(3.2) \quad K_0(u)^2 + K_0(v)^2 + K_0(u+v)^2 \geq e(H) := \frac{(\sqrt{5} - 2)^2}{(1 + 2\pi^2 H^2)^2}.$$

Proof. Using the Fourier transform of f_0 , we have

$$K_0(x) = \frac{1}{2\sqrt{2}\sin(1/\sqrt{2})} \left(\operatorname{sinc}((\sqrt{2} - 2\pi x)/2) + \operatorname{sinc}((\sqrt{2} + 2\pi x)/2) \right),$$

where $\operatorname{sinc}(x) = \sin x/x$ if $x \neq 0$ and $\operatorname{sinc}(0) = 1$.

Let $a = \sqrt{2}\pi \cot(1/\sqrt{2})$ and $\mathcal{F}(x) := ax \sin(\pi x) - \cos(\pi x)$. Combining the fractions gives

$$(3.3) \quad \mathcal{F}(x) = (2\pi^2 x^2 - 1)K_0(x).$$

Set $d = \max(|\mathcal{F}(u)|, |\mathcal{F}(v)|, |\mathcal{F}(u+v)|)$. We show $d \geq \sqrt{5} - 2$. It suffices to consider $d < 1$. Write $\alpha = au$, $\beta = av$, and $A = 1 + \alpha^2 + \alpha\beta + \beta^2$. Consider the vectors $X = (\sin(\pi u), \cos(\pi u))$, $Y = (\sin(\pi u), \alpha \sin(\pi u))$ in $\mathbb{R}^2$, equipped with the Euclidean norm $\|\cdot\|$. We have

$$\|X\| = 1, \quad \|Y\| = \sqrt{1 + \alpha^2} |\sin(\pi u)|,$$

and, by the definition of $\mathcal{F}$, $Y - X = (0, \alpha \sin(\pi u) - \cos(\pi u)) = (0, \mathcal{F}(u))$. The triangle inequality then gives

$$\left| \sqrt{1 + \alpha^2} |\sin(\pi u)| - 1 \right| = \left| \|Y\| - \|X\| \right| \leq \|Y - X\| = |\mathcal{F}(u)| \leq d.$$

Consequently,

$$1 - d \leq \sqrt{1 + \alpha^2} |\sin(\pi u)| \leq 1 + d.$$

Likewise, for v, β , we have

$$1 - d \leq \sqrt{1 + \beta^2} |\sin(\pi v)| \leq 1 + d.$$

By the trigonometric addition formulas,

$$\mathcal{F}(u+v) = A \sin(\pi u) \sin(\pi v) - \beta \sin(\pi u) \mathcal{F}(v) - \alpha \sin(\pi v) \mathcal{F}(u) - \mathcal{F}(u) \mathcal{F}(v).$$

Taking absolute values in this identity first gives

$$A |\sin(\pi u) \sin(\pi v)| \leq d + d^2 + d\beta |\sin(\pi u)| + d\alpha |\sin(\pi v)|.$$

Multiply by $\sqrt{(1 + \alpha^2)(1 + \beta^2)}/A$. Use the lower bounds for the two sine factors on the left and the upper bounds on the right. We obtain

$$(1 - d)^2 \leq d(1 + d) \frac{\sqrt{(1 + \alpha^2)(1 + \beta^2)} + \alpha\sqrt{1 + \alpha^2} + \beta\sqrt{1 + \beta^2}}{A}.$$

By using the inequality $t\sqrt{1 + t^2} \leq t^2 + 1/2$ for $t \geq 0$ and the arithmetic-geometric mean inequality $\sqrt{(1 + \alpha^2)(1 + \beta^2)} \leq 1 + (\alpha^2 + \beta^2)/2$ for $\alpha, \beta \geq 0$, the numerator is at most $2 + \frac{3}{2}(\alpha^2 + \beta^2) \leq 2A$. Hence $(1 - d)^2 \leq 2d(1 + d)$, or $d^2 + 4d - 1 \geq 0$, which implies $d \geq \sqrt{5} - 2$ for $d \geq 0$. Thus, by (3.3) at least one of the three values $K_0(u)$, $K_0(v)$, $K_0(u+v)$ has absolute value at least $(\sqrt{5} - 2)/(1 + 2\pi^2 H^2)$. Then (3.2) follows immediately by squaring. $\square$

4. PROOF OF THEOREM 1.1

We first cite a result of Lamzouri, which comes from [6, Lemma 3.2] and the proof therein. Let

$$C_{\text{MT}} := \frac{1}{2} + \frac{1}{\sqrt{2}} \cot(1/\sqrt{2}) = 2 - C_0.$$

Lemma 4.1 ([6, Lemma 3.2]). *For any $\varepsilon > 0$, there exists a real even function $\eta_\varepsilon \in C_c^\infty((-\frac{1}{2}, \frac{1}{2}))$ such that, with $f_\varepsilon = \eta_\varepsilon^2$, $K_\varepsilon = \widehat{f_\varepsilon}$ and $Q_\varepsilon = f_\varepsilon * f_\varepsilon$, we have*

$$\int_{-\infty}^{\infty} f_\varepsilon(u) du = 1, \quad f_\varepsilon \longrightarrow f_0 \quad \text{in } L^1(\mathbb{R}) \cap L^2(\mathbb{R}),$$

as $\varepsilon \rightarrow 0$, and for each fixed $\varepsilon > 0$, as $T \rightarrow \infty$, we have

$$(4.1) \quad \sum_{\substack{\rho, \rho' \\ 0 < \gamma, \gamma' \leq T}} K_\varepsilon \left(\frac{i(\rho - \rho') \log T}{2\pi} \right)^2 = (C_{\eta_\varepsilon} + o_\varepsilon(1))N(T),$$

where

$$C_{\eta_\varepsilon} = Q_\varepsilon(0) + 2 \int_0^1 \alpha Q_\varepsilon(\alpha) d\alpha \longrightarrow C_{\text{MT}},$$

as $\varepsilon \rightarrow 0$.

Now, we prove Theorem 1.1.

Proof of Theorem 1.1. For T large, let $\mathcal{Z}_T$ be the finite multiset defined by

$$\mathcal{Z}_T := \left\{ i \left(\rho - \frac{1}{2} \right) \frac{\log T}{2\pi} : 0 < \gamma \leq T \right\},$$

where every zero occurs with its multiplicity. The functional equation, through $\rho \mapsto 1 - \bar{\rho}$, makes $\mathcal{Z}_T$ invariant under complex conjugation. Its simple real elements correspond exactly to the zeros counted by $N_0^s(T)$. In the notation of Proposition 2.1,

$$n = N_0^s(T), \quad \sum_{z \in \mathcal{Z}_T} 1 = N(T).$$

By the Riemann–von Mangoldt formula (e.g., see [13]), the real elements lie in an interval of length $B_T := T \log T / (2\pi) = N(T) + o(N(T))$.

Fix $H > 0$; the symbol H denotes the cell length, leaving R for the set of real elements as in Lamzouri's proof. Partition the interval into $M_T = \lceil B_T/H \rceil$ cells of length at most H . In each cell form disjoint triples of simple real elements, leaving at most two unused. The total number J_T of triples satisfies

$$(4.2) \quad J_T \geq \frac{n - 2M_T}{3} \geq \frac{n - 2B_T/H}{3} - \frac{2}{3}.$$

Put $r_\varepsilon := \|f_\varepsilon - f_0\|_1 := \int_{\mathbb{R}} |f_\varepsilon(u) - f_0(u)| du$. On the real axis both kernels are real and bounded in modulus by one, and $\sup_{x \in \mathbb{R}} |K_\varepsilon(x) - K_0(x)| \leq r_\varepsilon$. For a triple $x < y < z$ in one cell, put $u = y - x$ and $v = z - y$. Then $u, v \geq 0$ and $u + v = z - x \leq H$. By Lemma 3.2,

$$K_0(u)^2 + K_0(v)^2 + K_0(u+v)^2 \geq e(H).$$

For every real t , by $|K_\varepsilon(t)|, |K_0(t)| \leq 1$, we have

$$|K_\varepsilon(t)^2 - K_0(t)^2| \leq |K_\varepsilon(t) - K_0(t)| (|K_\varepsilon(t)| + |K_0(t)|) \leq 2r_\varepsilon.$$

Since both kernels are real-valued on the real axis, this implies

$$K_\varepsilon(t)^2 \geq K_0(t)^2 - 2r_\varepsilon.$$

It follows that

$$K_\varepsilon(u)^2 + K_\varepsilon(v)^2 + K_\varepsilon(u+v)^2 \geq K_0(u)^2 + K_0(v)^2 + K_0(u+v)^2 - 6r_\varepsilon \geq e(H) - 6r_\varepsilon.$$

Put $e_\varepsilon := e(H) - 6r_\varepsilon$. For fixed H , we have $e(H) > 0$ and $r_\varepsilon \rightarrow 0$; hence $e_\varepsilon > 0$ for all sufficiently small ε .

Fix ε small enough that $0 < e_\varepsilon < 1/2$, and set $a_\varepsilon := 2e_\varepsilon/3$. By Lemma 3.1 and (4.2),

$$(4.3) \quad \Delta_{K_\varepsilon}(\mathcal{Z}_T) \geq 2e_\varepsilon J_T \geq a_\varepsilon \left(n - \frac{2N(T)}{H} \right) - o(N(T)).$$

Write

$$\mathcal{E}_{K_\varepsilon} = \sum_{z,s \in \mathcal{Z}_T} K_\varepsilon(z-s)^2.$$

Then by (2.1) of Proposition 2.1,

$$n \geq 2N(T) - \mathcal{E}_{K_\varepsilon} + \Delta_{K_\varepsilon}(\mathcal{Z}_T).$$

The analytic estimate (4.1) gives

$$(4.4) \quad \mathcal{E}_{K_\varepsilon} = (C_{\eta_\varepsilon} + o_\varepsilon(1))N(T),$$

while the preceding bound (4.3) for the spectral remainder yields

$$(4.5) \quad \Delta_{K_\varepsilon}(\mathcal{Z}_T) \geq a_\varepsilon \left(n - \frac{2N(T)}{H} \right) - o_\varepsilon(N(T)).$$

Substituting these estimates, we obtain

$$n \geq a_\varepsilon n + \left(2 - C_{\eta_\varepsilon} - \frac{2a_\varepsilon}{H} \right) N(T) - o_\varepsilon(N(T)).$$

This implies that

$$(4.6) \quad (1 - a_\varepsilon) \frac{N_0^s(T)}{N(T)} \geq 2 - C_{\eta_\varepsilon} - \frac{2a_\varepsilon}{H} - o_\varepsilon(1).$$

The error term tends to zero as $T \rightarrow \infty$.

Take $T \rightarrow \infty$ with H, ε fixed, and only then let $\varepsilon \rightarrow 0$. With $a_H := 2e(H)/3$, this yields

$$(4.7) \quad \liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq u_H := \frac{C_0 - 2a_H/H}{1 - a_H}.$$

By the definition of a_H , we have

$$u_H - C_0 = \frac{a_H(C_0 - 2/H)}{1 - a_H}.$$

Taking $H = H_0$ defined by (1.3) gives $a_H = a_0$ and $u_H = C_0 + \delta_0$, which proves (1.4).

For the distinct zero case, write $n = N_0^s(T)$. By (2.2) of Proposition 2.1,

$$N_d(T) \geq \frac{3}{2}N(T) - \frac{1}{2}\mathcal{E}_{K_\varepsilon} + \frac{1}{2}\Delta_{K_\varepsilon}(\mathcal{Z}_T).$$

Substituting the estimates (4.4) and (4.5) and dividing by $N(T)$, we obtain

$$\frac{N_d(T)}{N(T)} \geq \frac{3 - C_{\eta_\varepsilon}}{2} + \frac{a_\varepsilon}{2} \left(\frac{n}{N(T)} - \frac{2}{H} \right) - o_\varepsilon(1).$$

Fix H and a sufficiently small $\varepsilon > 0$, so that $0 < a_\varepsilon < 1$, and put

$$u_\varepsilon := \frac{2 - C_{\eta_\varepsilon} - 2a_\varepsilon/H}{1 - a_\varepsilon}.$$

Then the estimate (4.6) established for simple critical-line zeros gives

$$\liminf_{T \rightarrow \infty} \frac{n}{N(T)} \geq u_\varepsilon.$$

It follows that

$$\begin{aligned} \liminf_{T \rightarrow \infty} \frac{N_d(T)}{N(T)} &\geq \frac{3 - C_{\eta_\varepsilon}}{2} + \frac{a_\varepsilon}{2} \left(\liminf_{T \rightarrow \infty} \frac{n}{N(T)} - \frac{2}{H} \right) \\ &\geq \frac{3 - C_{\eta_\varepsilon}}{2} + \frac{a_\varepsilon}{2} \left(u_\varepsilon - \frac{2}{H} \right). \end{aligned}$$

Now let $\varepsilon \rightarrow 0$. We have

$$C_{\eta_\varepsilon} \rightarrow C_{\text{MT}} = 2 - C_0, \quad a_\varepsilon \rightarrow a_H := \frac{2e(H)}{3}, \quad u_\varepsilon \rightarrow u_H,$$

and hence

$$\liminf_{T \rightarrow \infty} \frac{N_d(T)}{N(T)} \geq \frac{1 + C_0 + a_H(u_H - 2/H)}{2} = \frac{1 + u_H}{2}.$$

Taking $H = H_0$, we have $u_H = C_0 + \delta_0$, $\frac{1+u_H}{2} = C_1 + \frac{\delta_0}{2}$. This proves (1.5). $\square$

Remark 4.2. In [14], by applying the inequality and method of Lamzouri, the author obtained explicit proportions of simple critical zeros and distinct zeros of the Riemann zeta function in short intervals. One can use the refined Lamzouri's inequality in Proposition 2.1 to improve these proportions.

ACKNOWLEDGMENTS

ChatGPT-6 Astra was used to discover the extra term $\Delta_K(\mathcal{Z})$ in Proposition 2.1 and implement the ideas of the proof of Theorem 1.1. The author verifies, corrects and rewrites the proof, and takes responsibility for the content.

REFERENCES

- [1] L. Alpöge and R. Furman, *More than two thirds of the zeta zeros are simple and on the critical line*, arXiv:2608.13637v2 (2026), <https://arxiv.org/abs/2608.13637v2>.
- [2] S. A. C. Baluyot, D. A. Goldston, A. I. Suriajaya and C. L. Turnage-Butterbaugh, *An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta function*, Acta Arith. **214** (2024), 357–376.
- [3] J. B. Conrey, *More than two fifths of the zeros of the Riemann zeta function are on the critical line*, J. Reine Angew. Math. **399** (1989), 1–26.
- [4] G. H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, C. R. Acad. Sci. Paris **158** (1914), 1012–1014.
- [5] S. Feng, *Zeros of the Riemann zeta function on the critical line*, J. Number Theory **132** (2012), no. 4, 511–542.
- [6] Y. Lamzouri, *A new proof that more than 2/3 of the zeros of the Riemann zeta function are simple and on the critical line*, arXiv:2609.02882v2 (2026), <https://arxiv.org/abs/2609.02882v2>.
- [7] N. Levinson, *More than one third of zeros of Riemann's zeta-function are on $\sigma = 1/2$* , Adv. Math. **13** (1974), 383–436.
- [8] S. H. Min, *On the non-trivial zeros of the Riemann zeta function*, Acta scientiarum naturalium Universitatis Pekinensis **2**(2) (1956), 165–189.
- [9] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Sympos. Pure Math. **24** (1973), 181–193.
- [10] H. L. Montgomery, *Distribution of the zeros of the Riemann zeta function*, Proceedings of the International Congress of Mathematicians (Vancouver, 1974), Vol. 1, Canadian Mathematical Congress, 1975, 379–381.
- [11] K. Pratt, N. Robles, A. Zaharescu and D. Zeindler, *More than five-twelfths of the zeros of ζ are on the critical line*, Res. Math. Sci. **7** (2020), Paper No. 2.
- [12] A. Selberg, *On the zeros of Riemann's zeta-function*, Skr. Norske Vid. Akad. Oslo I **10** (1942), 1–59.
- [13] E. C. Titchmarsh, *The Theory of the Riemann Zeta-function*, second edition, revised by D. R. Heath-Brown, Oxford University Press, 1986.
- [14] B. Wang, *Simple critical zeros and distinct zeros of the Riemann zeta-function in short intervals*, arXiv:2609.07918 (2026), <https://arxiv.org/abs/2609.07918>.

SCHOOL OF MATHEMATICS AND STATISTICS, YUNNAN UNIVERSITY, KUNMING, YUNNAN 650500, CHINA
Email address: bwang@ynu.edu.cn